

TOC (CS302)

Complete Syllabus

Course Overview

Theory of Computation introduces the mathematical foundations of computer science, focusing on formal models of computation and language recognition.

Course Objectives

Understand automata theory, formal languages, computability, and complexity concepts used in compiler design, artificial intelligence, and algorithm analysis.

Learning Outcomes

Students will be able to design finite automata, construct grammars, analyze language classes, and evaluate the computability of problems.

Prerequisites

Discrete Mathematics, Data Structures, and basic knowledge of algorithms.